

Automatic WATER LEVEL INDICATOR Cum CONTROLLER

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Water storage often presents challenges such as monitoring tank levels, preventing overflow, and reducing

manual intervention to operate the pump. To address these issues, two types of devices, known as water level indicators and water level controllers, have been developed.

These devices have key drawbacks. Water level indicators require manual pump operation and often

waste power as multiple LEDs stay unnecessarily lit. Water level controllers face frequent pump cycling from minor fluctuations, which accelerates the pump's wear and tear and shortens its lifespan.

An automatic water level indicator cum controller device can

TABLE 1
LIST OF COMPONENTS

Item	Quantity
Arduino Uno (MOD1)	1
Water level sensors or float switch	3
5V SPDT relay (RL1)	1
BC547 NPN transistor (T1)	1
1N4007 rectifier diode (D1)	1
330Ω resistor (R1, R2, R3)	3
10kΩ resistor (R4, R5, R6)	3
1kΩ resistor (R7)	1
Water pump	1
5mm LED	3
Power supply for Arduino Uno	5V
Power supply for water pump motor	230V AC

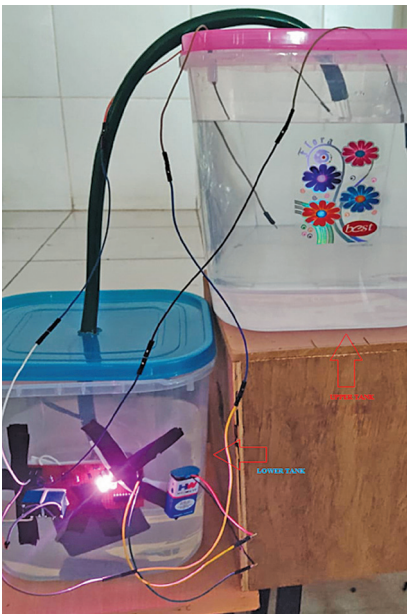


Fig. 1: Author's prototype

etch_jan15a.ino

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1 //RAKESH JAIN PROGRAM FOR FULLY AUTOMATIC WATER LEVEL INDICATOR CUM CONTROLLER
2 const int sensorL1 = 2; // Lowest sensor
3 const int sensorL2 = 3; // Middle sensor
4 const int sensorL3 = 4; // Highest sensor
5 const int LEDL1 = 8; // Lowest level LED
6 const int LEDL2 = 9; // Middle level LED
7 const int LEDL3 = 10; // Highest level LED
8 const int relayPin = 11; // Relay control pin
9 // Pump state variable
10 bool pumpOn = false; // Tracks whether the pump should be ON or OFF
11 void setup() {
12 // Configure sensor pins as input
13 pinMode(sensorL1, INPUT);
14 pinMode(sensorL2, INPUT);
15 pinMode(sensorL3, INPUT);
16 // Configure LED and relay pins as output
17 pinMode(LEDL1, OUTPUT);
18 pinMode(LEDL2, OUTPUT);
19 pinMode(LEDL3, OUTPUT);
20 pinMode(relayPin, OUTPUT);
21 // Initialize the relay (pump OFF)
22 digitalWrite(relayPin, LOW);
23 }
24 void loop() {
25 // Read sensor values
26 bool levelL1 = digitalRead(sensorL1);
27 bool levelL2 = digitalRead(sensorL2);
28 bool levelL3 = digitalRead(sensorL3);
29 // Filling Logic
30 if (!levelL1 && !levelL2 && !levelL3) {
31 // Water below lowest sensor
32 pumpOn = true; // Turn pump ON
33 }
34 else if (levelL3) {
35 // Water reaches highest sensor
36 pumpOn = false; // Turn pump OFF
37 }
38 // Update Relay State
39 digitalWrite(relayPin, pumpOn ? HIGH : LOW);
40 // LED Indicators
41 if (levelL1 && !levelL2 && !levelL3) {
42 // Water touches lowest sensor
43 digitalWrite(LEDL1, HIGH); // LED 1 ON

```

Fig. 2: Snippet of source code